

# The Autonomous Whiteboard Cleaner

## ME 397

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### **Abstract**

This project focuses on developing a compact, detachable, and low-cost autonomous whiteboard cleaning robot to improve efficiency and convenience in everyday environments. The primary problem addressed is the lack of an affordable, user-friendly system that automatically cleans vertical whiteboards while maintaining safe, reliable operation. Our solution was to design a robot that moves across the whiteboard in a strategic pattern, like a robotic vacuum, using magnetic adhesion to overcome gravity, and keep in contact with the surface.

The system was constructed using a fully 3D-printed frame made from PETG, designed to securely house all components while staying lightweight and durable. Custom wheels were developed to fit precisely onto the DC motor shafts, with added rubber for improved traction and stability. The robot is powered by a 12V battery, with a step-down buck converter supplying proper voltage to the microcontroller and sensors. An integrated control system uses IR sensors and motor encoders to detect edges and track motion, enabling the robot to navigate and clean effectively with a felt pad.

Throughout development, key challenges included structural failures, magnetic force balancing, and control system implementation. Despite these obstacles, the final design shows a functional, portable, and cost-effective solution, setting up a solid foundation for future improvements and potential commercial applications.

### **Introduction**

In classrooms and offices, automation has become more common to address productivity breaking and repetitive tasks. Hand erasing whiteboards can take 5 to 10 minutes per class. This leads to lost instruction or work time over the course of the year. Existing automated solutions are either too costly, are not portable, produce too much noise, or require permanent installation like fixed-rail cleaners and rack-and-pinion systems. These constraints have created obvious demand for a compact, low cost, and portable system. From which cleans a whiteboard without major infrastructure changes. Improving efficiency by reducing manual work in everyday tasks. One common repetitive task is cleaning whiteboards in classrooms and offices, which interrupts workflow and lowers productivity. This project focuses on developing an autonomous whiteboard-cleaning system to provide a more efficient and convenient solution.

The goal is to design a compact, low-cost, portable robot that works autonomously and efficiently, while staying within a \$100 budget. The system is lightweight, detachable, and designed to function on both horizontal and vertical surfaces using magnetic adhesion. The design process involved multiple iterations. CAD models were used to plan the layout, followed by early cardboard prototypes to test size and spacing. The final design was built using a 3D-

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printed PETG frame. The system integrates an Arduino, motor driver, DC motors with encoders, IR sensors, and a buck converter. Programming enabled motor control and basic autonomous behavior, including obstacle detection.

During testing, challenges such as low motor power, unstable power supply, weak magnets, and shaft failures were found. These led to improvements including a simplified box frame, better weight distribution, stronger magnets, and a redesigned wheel system with an integrated shaft. The robot showed movement and basic functionality on both surfaces, though further improvements are needed for consistent vertical climbing. This project highlights the importance of iterative design and problem-solving in engineering.

## Literature Review

Automation is becoming more common because it helps save time and reduce manual work. In classrooms, teachers spend about 5–10 minutes per class erasing whiteboards, which can add up to over 30 hours per year. This interrupts lessons and reduces teaching time, so an autonomous whiteboard cleaner could be useful for teachers, students, and professionals. Previous designs include fixed-rail systems [1], which move along a mounted track; rack-and-pinion mechanisms [2], which use gears to drive motion across the board; brush-based cleaners [3], which use rotating brushes to wipe the surface; fixed-mounted dusters [4], which stay in one position and clean a limited area; and guided robots [5], which require a laser to guide the robot in the direction it should clean. While these systems improve cleaning consistency and reduce effort, they often lack portability, create noise, require permanent installation, or are too expensive for everyday use.

For example, fixed systems [1], [2] are not adaptable to different board sizes, while simpler designs [3], [4] are more affordable but lack flexibility and control. More advanced systems like the guided robot [5] improve accuracy but require extra setup and increase cost and complexity beyond our budget. Overall, these limitations show the need for a design that balances performance, portability, low noise, and affordability. This project focuses on developing a compact and cost-effective solution that can be easily used in real classroom and office environments

## Safety codes and standards

The autonomous whiteboard cleaner is designed to meet important robotic and electrical safety standards. This ensures safe and reliable operation in classrooms and offices. Standards like Robotic Industries Association TR R15.306 and ISO 10218 help with finding hazards, reducing risks, and setting up emergency stop systems. Human-robot interaction requirements from ISO/TS 15066 set safe speed and force limits for work around users. Electrical and

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mechanical safety follow IEC standards and CSA Group codes, including CSA C22.2 No. 336. These define the requirements for proper wiring, grounding, insulation, and safe integration of components. Functional safety guidelines from IEC 63327 require systems to detect obstacles, prevent collisions, and shut down safely. Added standards like ANSI/RIA R15.06 and ISO 13482 ensure safe operation in areas with non-expert users. These emphasize clear labeling, enclosed components, safe startup procedures, and controls that users can access. Together, these standards, summarized in Table 1, guide the overall design. This includes the placement of emergency stops, sensor choice, enclosure protection, electrical safety, and system reliability.

Table 1 – Codes and Standards that Apply to the Design of a Whiteboard Eraser

| Subject Matter                            | Code                                             | Contents                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Safety</b>                     | RIA TR R15.306                                   | hazard Identification and mitigation                                                                                                                                                                                                                                                                                                 |
|                                           | ISO 10218-1/-2                                   | Emergency Stop Functionality                                                                                                                                                                                                                                                                                                         |
|                                           | ANSI/A3 R15.06                                   |                                                                                                                                                                                                                                                                                                                                      |
|                                           | ISO/TS 15066                                     | Safe Speed & Force Limits                                                                                                                                                                                                                                                                                                            |
| <b>Electrical &amp; Mechanical Safety</b> | <b>ISO 18650</b>                                 | Electrical Compliance                                                                                                                                                                                                                                                                                                                |
|                                           | IEC and CSA                                      | Insulation & Grounding Verified                                                                                                                                                                                                                                                                                                      |
| <b>Functional Safety</b>                  | <b>IEC 63327 and CSA 22.2 No. 336</b>            | Obstacle Detection & Avoidance, Collision Prevention, and Fail-Safe Mode                                                                                                                                                                                                                                                             |
| <b>Software &amp; Cybersecurity</b>       | <b>ISO 10218-2</b>                               | Control System Safety / Cybersecurity Measures and Data Privacy                                                                                                                                                                                                                                                                      |
| <b>Human Interaction</b>                  | ISO/TS 15066                                     | Collaborative Operations Risk Analysis, Clear User Instructions, and Accessible Emergency Stop                                                                                                                                                                                                                                       |
| <b>Regulatory Compliance</b>              | <b>ANSI/A3 R15.06 (U.S.)</b>                     | All safety features must be implemented and documented. The robot must have clear labelling, safe startup procedures, and user-accessible controls. If used in schools or offices, it must be secure for non-expert users.                                                                                                           |
|                                           | <b>ISO 10218 &amp; ISO 13482 (International)</b> | Emergency Stop Buttons, Avoid sharp edges, securely enclosed, safe unsupervised use, Low noise level, easy to clean                                                                                                                                                                                                                  |
|                                           | <b>CSA 22.2 No. 336 (Canada)</b>                 | The robot must have obstacle detection sensors, it must navigate safely around furniture, walls, and people, it must not exceed safe operating speeds and must stop if lifted or tilted                                                                                                                                              |
|                                           | <b>IEC 63327 (Global)</b>                        | If the robot uses liquids for cleaning, it must have waterproofing or splash protection for internal electronics. All electrical components must be adequately insulated and grounded to prevent electric shock. Must include instructions for safe operation, maintenance, and troubleshooting. Physical labels must say hazardous. |

### Problem Statement and objectives

The aim of this project was to design an autonomous whiteboard cleaning robot that runs without being permanently attached to the surface while keeping a total cost of approximately \$100 or less. A primary challenge was deciding how the robot would adhere to the whiteboard without a fixed mounting system; after evaluating multiple approaches, magnetic adhesion was

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selected due to its simplicity, low cost, and reliable performance. In terms of overall design, the team adopted a concept inspired by a Roomba, changing its autonomous navigation strategy for vertical surface cleaning instead of floor operation, allowing for efficient and consistent coverage of the whiteboard. Additionally, careful consideration was given to part selection, including the microcontroller, motors, sensors, wheels, battery, and chassis materials, all chosen to balance performance with cost constraints. The system is controlled using an Arduino Uno R3, which was selected for its affordability, ease of use, and extensive availability of online resources.

### **Design Requirements and Constraints**

The autonomous whiteboard cleaner is designed around the needs of a classroom setting and users who require a low cost, portable, and easy to use for cleaning and erasing on a whiteboard. The design requirements were made and developed from the sponsors' expectations. Which consisted of budget limits, safety considerations, and working with limitations on a vertical whiteboard. Using these set requirements and goals given to the team, it helped guide the teams' selections of its frame, adhesion system, sensors, cleaning method, power source, control hardware, and drive system.

The most important design constraints were the cost and a reliable adhesion system. Since the robot would work on a vertical surface, the magnetic attachment system must provide a reliable holding force to prevent the robot from falling. This magnetic attachment must also prevent any slipping that would cause it to fall, all while allowing the wheels to have contact on the surface and move across the board. The robot must also avoid scratching the surface of the board, be able to run quietly in a classroom or office environment, remain simple enough to be made and manufactured all while keeping within the projects budget limit.

The primary constraint for this project would be cost. Although the maximum budget for this project was \$500, the goal was to keep it under or around \$100 range in total to manufacture and meet the requirements. This constraint limits the project by potentially adding different sensors such as color sensors or LiDAR sensors as examples. Or several types of DC motors or rechargeable batteries would be another set of upgrades that could make the robot run better and more efficient. But with such constraints, this led the team to make lower cost selections of components and work on it to be as simple and efficient as possible with the set budget constraints given to the team.

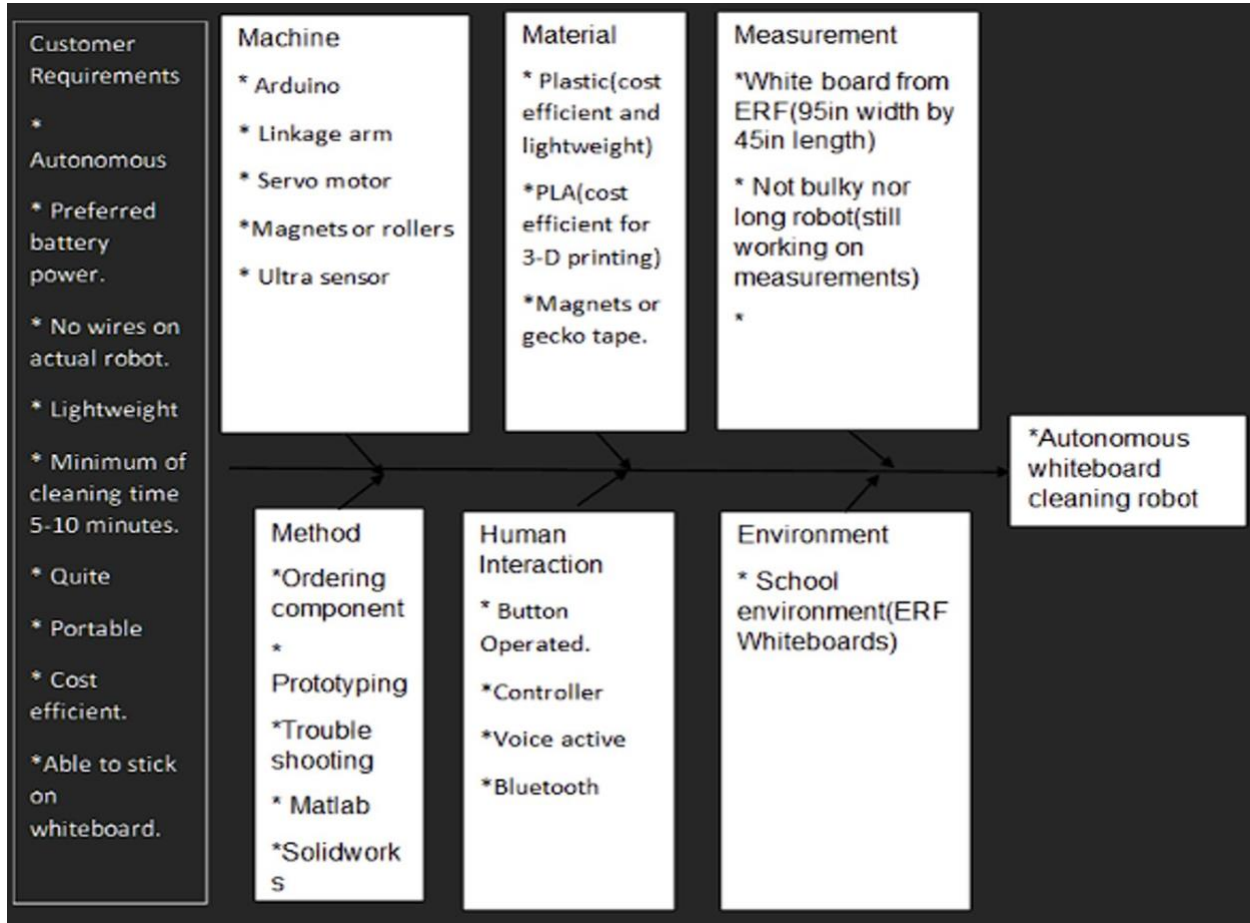
## **6. Proposed Solution & Alternatives**

This section presents the design alternatives considered before the final prototype development, the metrics used to compare them, and the final proposed solution selected from the decision matrix.

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## Concepts

### Fishbone Diagram



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A morphological chart was created to explore unique design options for each major part of the autonomous whiteboard cleaner. The chart helped organize solutions for key functions like adhesion, movement, cleaning method, power supply, sensing, control, navigation, and safety. For each function, we considered multiple options, like magnetic adhesion, different chassis designs, drivetrain setups, and cleaning methods such as rotating brushes or microfiber pads.

This approach allowed us to compare each possibility based on performance, cost, portability, and how well it meets the safety standards listed in Table 2. For example, more advanced sensing options like cameras or LiDAR would give better accuracy, but they were too expensive for our budget, so we focused on simpler options like ultrasonic sensors. We also looked at movement systems like ceiling rails or suction, but these were not practical because they reduced portability. Overall, using the morphological chart helped us narrow down the best combination of components that meet our goals, including portability, low cost, quiet operation, and safe performance. Table 1- Morph Chart

| Functions              | Mean 1                   | Mean 2                                     | Mean 3                    | Mean 4                | Mean 5                  | Mean 6             | Mean 7                 | Mean 8              |
|------------------------|--------------------------|--------------------------------------------|---------------------------|-----------------------|-------------------------|--------------------|------------------------|---------------------|
| <b>Adhesion</b>        | Vacuum suction pump      | Magnets                                    | Sticky polymer strip      |                       |                         |                    |                        |                     |
| <b>Body/Frame</b>      | RC Car (Piggyback route) | 3D Printed (Exact dimensions TBD (4.5 x 7) |                           |                       |                         |                    |                        |                     |
| <b>Movement Method</b> | Magnetic drive           | Wheels                                     | Timing based grid pattern | Bumper Switches       | Omni wheel              | Stepper motor      | Suction crawler tracks | Ceiling rail gantry |
| <b>Cleaning Method</b> | Rotating brush/eraser    | Microfiber                                 | Eraser Bar                | Spray and wipe System | Vibration clearing pads |                    |                        |                     |
| <b>Power</b>           | Rechargeable Battery     | Replaceable Battery                        |                           |                       |                         |                    |                        |                     |
| <b>Sensor</b>          | Bumper Switches          | Camera Detection                           | Ultrasonic detection      | Hall Effect sensor    | IR System detection     | Lidar micro-Module |                        |                     |
| <b>Control</b>         | Arduino                  | Raspberry Pi                               | ESP32                     | Bluetooth             |                         |                    |                        |                     |
| <b>Navigation</b>      | Random bounce pattern    | Grid path                                  | Edge follow algorithm     | Marker tracker        |                         |                    |                        |                     |
| <b>Safety</b>          | Emergency shutoff switch | Led indicator                              | Soft bumpers              |                       |                         |                    |                        |                     |

## Alternative concepts

Concept 1: Magnetic IR Sensor Supported Robotic (Design Selected).

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The first concept is a mobile robotic vehicle, which uses permanent magnets to adhere itself to the whiteboard, and then uses infrared (IR) sensors to travel over the surface. IR sensors identify differences in reflectivity from the whiteboard to marker ink, so the robot can tell which areas are at or needing cleaning. Motion is done by DC motors and wheels, while an Arduino microcontroller processes sensor data and controls movement. The design was chosen, especially as the system was easy to produce and perform, requiring no advanced computation or costly devices. The design is intended to be portable and detachable, and means it does not require rails, or a suction based system. The robot follows the idea of a small autonomous cleaning device, similar to a robotic vacuum, but adapted for vertical whiteboard cleaning.

The system for adhesion uses magnets placed near the wheels and center of mass to keep the robot attached to the board. The placement of the magnets is important because the robot needs enough holding force (mentioned in engineering analysis) to prevent any slipping, but not so much force that the motors could not move it. The rubber wheels improve traction and help move the robot across the board with no problems. Motion is allowed through the use of 2 DC gear motors controlled and powered by an Arduino through a motor driver. The Robot uses IR sensors to detect any edges of the whiteboard and changes in surfaces. A ToF sensor in the front is used to detect any obstacles in the robot's path and if anyone's hand is in the way (safety mechanism). Based on the sensor readings, the Arduino would stop, turn, and go to continue cleaning on the board. Like a green, yellow, red system. Detects the board so it's safe to move, detects something on one sensor, so slightly turn, Or it detects nothing or something blocking the way, stop immediately and turn around.

The cleaning system incorporates microfiber or felt underneath the chassis. As the robot moves, the pad would come in contact with the board and erase what was left from the markers. The pad would also help add friction (explained in engineering analysis) and a layer between the magnets and the board, which creates an effective magnetic pull that's great to stay on vertical surfaces and clean effectively. Because of this, it keeps the cleaning method simple and low cost.

### Concept 2: Suction-Based Mobile Whiteboard Cleaner

A suction-based robot that uses a vacuum or fan system to adhere to the vertical whiteboard surface instead of magnets. This design would use a small vacuum fan or suction



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motor to create negative pressure between the robot and the whiteboard, allowing it to remain attached while moving across the board. A flexible rubber or foam seal would be placed around the bottom of the chassis to help maintain suction and reduce air leakage. The robot would move using DC motors and rubber wheels, while a microfiber or felt cleaning pad would erase marker residue as the robot travels across the surface.

This concept has the advantage of being usable on many types of whiteboards, including boards that are not magnetic. Since the adhesion force would come from suction rather than magnets, the robot would not depend on the whiteboard having a steel backing. This could make the design more versatile in classrooms and offices. The suction force could also be adjusted by changing the fan speed, allowing the robot to balance adhesion and mobility more easily than a fixed magnet system.

However, this concept also has several disadvantages. The suction system would require an additional motor or fan, which increases cost, weight, power consumption, and noise. Since the robot is intended for classroom or office use, the added noise from the vacuum fan could be distracting. The extra power draw would also reduce battery life and require a larger battery, making the robot heavier and more difficult to keep attached to the board. Another major issue is maintaining a reliable seal against the whiteboard surface. If the seal is not smooth or if the board has gaps, marker residue, frame edges, or surface imperfections, the robot could lose suction and fall.

This design was not selected because it was more complex and less practical for the project constraints. Although it would work on non-magnetic whiteboards, the added fan, sealing system, and larger power supply would make the design harder to manufacture within the budget. It would also introduce safety concerns because a sudden loss of suction could cause the robot to detach from the board. Compared with the magnetic adhesion concept, the suction-based robot requires more energy, creates more noise, and has more failure points. For these reasons, the suction-based concept was rejected in favor of the simpler magnetic mobile robot design.

### Concept 3: Magnetic Grid with Hall-Effect Navigation

The third idea uses a magnetic grid system found behind the whiteboard surface, coupled with Hall effect sensors to find position. The robot would have preconfigured magnetic paths that guided it, for a controlled and repeatable way of travelling around the area. This concept has the advantage of improving navigation accuracy compared with basic edge detection. Since the robot would follow a known grid pattern, it could cover the board more systematically and reduce missed areas. It could also make the control logic easier once the grid system is installed.

Although this technique offers a systematic approach to movement and precision, we rejected this method because it must be adjusted to the whiteboard environment, which contradicts the project requirement for the robot to run without having to be permanently mounted or to require

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some other hardware. This concept was also not chosen because it requires modifying the whiteboard or adding external hardware. This goes against the project's goal of making the robot portable, detachable, and usable on different boards without permanent installation. It would also increase setup time, cost, and complexity. For these reasons, the magnetic grid concept was rejected.

### Concept 4: Camera Based Magnetic Robot with ESP32 Control

This concept design features a square, 3D-printed body. The robot utilizes a simple grid trajectory, which is controlled by a small camera connected to an ESP32 microcontroller. The system can move independently during scrubbing and spraying. The robot uses magnets to stick to the whiteboard, which is a good approach, but it also has some weaknesses. Potentially, when it is near the edges, with corners becoming sharp if it changes direction too quickly, this could result in separation when the friction force exerted by the wheels exceeds the magnetic force. Essentially, the risk of separation is always present because the magnetic force is dependent on contact. When the robot hits a wall head-on, it often takes too long to realize and prevent a fall.

This concept has the advantage of giving the robot more advanced sensing capability. With camera feedback, the robot could identify areas with writing and focus cleaning on those sections instead of moving blindly across the board. This could improve cleaning efficiency and reduce unnecessary movement.

However, this concept was not selected because it adds too much complexity for the project requirements. Camera processing would require more advanced programming and calibration than IR or ToF sensors. The spray system would also introduce safety concerns because liquid near electronics would require waterproofing and extra protection. In addition, the square body could make turning near the board edges more difficult and increase the risk of the robot losing contact with the surface. For these reasons, this concept was rejected in favor of a simpler magnetic robot with IR/ToF sensing.

## Concept Selection

Decision matrix

|               | Durability | Weight | Speed | Cost | Accuracy | Rating |
|---------------|------------|--------|-------|------|----------|--------|
| Weight Factor | 0.2        | 0.25   | 0.05  | 0.3  | 0.2      | 0/5    |

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|                        |     |      |      |     |     |             |
|------------------------|-----|------|------|-----|-----|-------------|
| Magnetic Micro cleaner | 4   | 5    | 2    | 4   | 3   | <b>3.95</b> |
|                        | 0.8 | 1.25 | 0.1  | 1.2 | 0.6 |             |
| 'High-tech' Navigation | 4   | 3    | 3    | 3   | 4   | <b>3.4</b>  |
|                        | 0.8 | 0.75 | 0.15 | 0.9 | 0.8 |             |
| Hall Effect Sensor     | 3   | 4    | 2    | 3   | 2   | <b>3</b>    |
|                        | 0.6 | 1    | 0.1  | 0.9 | 0.4 |             |
| Suction Based          | 3   | 2    | 3    | 2   | 3   | <b>2.45</b> |
|                        | 0.6 | 0.5  | 0.15 | 0.6 | 0.6 |             |
| Camera Based           | 3   | 3    | 3    | 2   | 5   | <b>3.10</b> |
|                        | 0.6 | 0.75 | 0.15 | 0.6 | 0.9 |             |

## Final Selection

Based on the descion matrix, Concept 1: Magnetic differential drive with IR sensor Robot design was selected as the final design. This concept received the highest weighted score becasue it provided the best balance of portability, cost, manufacturability, safety, and autonomous function. Unlike the Suction design, it offers a more reliable system of sticking onto the vertical wall that is easy to attach. As well as a simpler and cheaper option. Having a suction-based design for sticking onto the vertical surface would have created more problems than answers, such as how it would clean since that's the point of this design. Concept 1 also avoided using more expensive options for sensors that would introduce a more complex system to integrate into the design, sush as mapping out a grid map of the white board or simply using a color sensor. The IR sensors offer a simpler and more reliable system of moving around the whiteboard. Compared to concept 2, the selected design also avoids added friction and surface scratching risk caused by a castor wheel.

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The final proposed solution uses a PETG 3D-printed chassis, N20 DC gear motors, rubber wheels, permanent magnets, IR edge sensors, a front ToF obstacle sensor, an Arduino microcontroller, a TB6612FNG motor driver, a 12 V rechargeable battery, a buck converter, and a felt cleaning pad. This design was chosen because it best satisfies the main project goals: low cost, portability, safe vertical adhesion, autonomous movement, and effective whiteboard cleaning.

### **Methods and Results**

The design process was based on an engineering approach, where each prototype was developed, tested, evaluated, and updated considering their outcome. The main subsystems of the autonomous whiteboard cleaner consisted of chassis/structure, motion and adhesion, sensing/control, and power/electronics. Each design iteration was designed to refine and improve them to satisfy the design quality requirements of low cost, portability, autonomous operation, and reliable whiteboard cleaning. The first prototype allowed us to check whether the robot could be built to attach to a vertical surface and move across the board. Later prototypes also investigated areas such as strength, stability, motor placement, magnet placement, wheel traction, sensor reliability, and integration of components. This enabled the team to focus on testing and engineering decisions that were not purely theoretical. The final product was a functioning autonomous whiteboard cleaner which employed magnetic adhesion, motor-driven wheels, IR/ToF sensing, encoder feedback, and a cleaning pad to traverse and erase a whiteboard. This progress is shown in the project presentation, during the various prototype stages, the CAD/body concepts, motor and sensor testing, magnet placement revisions, and final frame improvements.

#### **- Chassis and Structural Design**

The chassis was one of the first big systems developed because it was a framework for how all the other elements would be assembled. The first design consisted of a 3D-printed frame

that was meant to house the motors, sensors, magnets, Arduino, wiring, and cleaning surface. Early concepts featured rounded and square shape frames based on designs in the space of a small robotic vacuum. Our aim was to keep the body small while providing the space for electronics as well as keeping the robot lightweight. Prototype No. 1 showed that the main part could be accommodated with a single frame, but testing showed that the design could also use a slight ergonomic adjustment. There wasn't enough real space inside the body to include the motors and wiring, and the placement of the motors prevented the wheels from fitting perfectly to each other. The next design improved it to an approximately 7 in  $\times$  5 in frame size, moved the wheels outward, and expanded the IR sensors' openings. This helped to move the internal layout to a better vantage point, as well as allowing the team to place the drive components better. More testing revealed that the frame still required simplifying. Certain earlier configurations were too bulky, while others caused misalignments between the motor shafts and wheels. The team also noticed that under load the shafts and connection points failed to resist, leading to misalignment and lower movement reliability. This then guided a shift towards building its chassis using a simplified box style that proved easier to print, construct, and maintain. Final design incorporated PETG 3D-printed chassis instead of PLA. PETG was selected for its superior durability and strength: it wasn't just in prototypes of an earlier round that weak parts were produced. The final frame with screws used both top and bottom plate designs to split the layers. This allowed the magnets, wheels, sensors, and cleaning pads to sit in alignment with each other and keep the robot level against the whiteboard. This layout helped stabilize, assemble, and form the overall structure in a reliable manner for the final device. A configuration also makes it easier to assemble, support, and replace some of the components.

### - **Motion and Adhesion System**

The motion and adhesion system were most important in this project, as the robot had to move around a vertical whiteboard without falling. The robot was mounted to the board using magnets embedded in the frame to form a tether, and motion was provided by DC motors and wheels in the preliminary design. This arrangement was selected to prevent permanent mounting and to allow for a portable system. Early tests showed that the idea was practical since the robot would be able to latch to a vertical surface. But early tests also suggested that the positioning of magnets significantly affected their performance. If the magnets were weak or not positioned effectively enough, the robot would slide or lose contact. If magnetic force was excessive, the wheels had a tough time overcoming the accumulated resistance. This led to a trade-off between adhesion and mobility in engineering form. The following version evaluated stronger magnets and different placement arrangements. The team tried bigger magnets around the wheels and smaller magnets around

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the frame. This made the robot have better attachment performance when in motion. The final prototype employed powerful magnets placed close to the drive wheels and the center of mass to distribute the weight of the robot and to sustain stable contact with the whiteboard. The presentation says they'd evaluated combinations of large magnets against smaller support magnets to enhance adhesion, and in time the adhesion was improved enough to accommodate vertical usage. Through this project, the drive system changed too. Early motors didn't give sufficient reliable movement during loading, which in part can only happen because the robot was attached magnetically to the board. The team swapped in new motors, tougher and slower; that added torque to the motors. That was a critical engineering decision as vertical movements were torque/drive-dependent, not speed-dependent. The wheel design also changed. Early wheel and shaft connections were weak and had alignment faults. Shafts broke, and wheels became angled while they were being tested, which led to reduced power transfer. The team stepped away from the original shaft configuration to develop custom wheels/adapters to fit the motor shafts more comfortably. The added rubber increased traction and decreased slipping. A caster ball/wheel was added to enhance stability and motion consistency. These modifications provide a more cohesive motion framework that enables the robot to traverse the surface of the system with greater efficiency.

### - Sensing and Control System

The power and electronics system were programmed to feed the motors, sensors, and controller while keeping the robot small and simple but also compact and portable. The first electronics system was Arduino, breadboard, wires, DC motors, motor driver, and battery. With this setup, the team was able to test fundamental motor control and sensor response before going ahead to finish the mechanical setup. Power delivery appeared critical early on when testing the system since motors needed current to carry the robot under magnetic load. The robot was powered by a battery, and the lower-voltage electronics were protected by a voltage regulation system. A buck converter was attached to reduce voltage for the Arduino and sensors, while the motor driver provided DC motors. The wiring and installation of part placement were of importance with subsequent design changes. Earlier prototypes had exposed wiring and temporary connections, some of them usable for testing but not great for the purpose of final use. Later versions were designed to organize wiring, tie down boards and battery, and situate elements inside the chassis to minimize movement interference. The final electronics design added the controller, motor driver, sensors, battery, and voltage regulation modules found within the frame. That produced a cleaner and more reliable system and kept the robot detachable and portable.

### - Power and Electronics System

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The power and electronics system were constructed to deliver the motors, sensors, and controller while minimizing the footprint of the robot and making it portable. The first electronics arrangement provided an Arduino, breadboard, wires, DC motors, motor driver, and battery. The experimental setup enabled the team to develop preliminary motor control and sensor response before completing the mechanical design. Early tests revealed that power supply of motors was crucial, as they required sufficient current to move the robot under a magnetic load. It is supported by a battery that fueled the robot as well as a voltage regulation system protecting the lower-voltage electronics. A buck converter was added to reduce the voltage for the Arduino and sensors, and a motor driver was used to power DC motors. Throughout the design process, wiring and location of components were increasingly important. Earlier prototypes had exposed wiring and provisional connections, good for experimenting but not well-suited for running full-blown operation. Later versions concentrated on fitting the wiring, adding screws and securing the boards and battery, and placing elements in the chassis to minimize interference with movement. The final electronics design featured the controller, motor driver, sensors, battery, and voltage regulation components mounted within the frame. This ended up in a cleaner and more reliable system to work with while also allowing this robot to be detachable and portable.

### **Final Design Iteration and Engineering Justification**

The final design itself resulted from a series of prototype investigations and engineering tradeoffs. The team started with a relatively simple concept that tested whether a magnetic robot could attach to and move on a whiteboard. The following design shift was built with that in mind.

Most of the design features were better as follows:

The frame was simplified into PETG for chassis evolution from bulky early. The magnet was reorganized to the right scale between adhesion and mobility. The motors were converted to higher, stronger motors. The wheel system is redesigned according to the design to improve traction and remove weak shaft failures via the entire wheel. The IR/ToF sensors had been reconfigured and calibrated for autonomous navigation support while IR/ToF sensors were implemented to be placed and adjustable for autonomous navigation. The electronics were rearranged for better reliability and to fit within the frame.

These changes show that the design is not merely random but evolves through experimentation and engineering judgment. The final device fulfills the primary goals of the project: by moving on a whiteboard autonomously, staying adhered to the surface via magnetic adhesion, perceiving external edges/obstacles, and wiping the surface with a passive cleaning pad. Future work could be more in-depth, but they are not changing to the current system because they are also improvements on top and not modifications to the original one, in the form of color detector capability, line tracking and more advanced path planning.

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### Prototype and Assembly

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After selecting the concept design that was a realistic path and aligned with what the team was looking for. This selected design was selected and used to develop a functional autonomous whiteboard cleaning robot prototype. This prototype was designed and used to show the basic functionalities and key functions of the system it would use. Which consists of magnet attachments, autonomous movements/edge detection, and the eraser. The final prototype consisted of a **3D printed frame**, two **N20 DC motors**, 2 rubber wheels, three **IR sensors**, an **Arduino** control system, **motor driver**, a buck converter, a rechargeable battery, magnets, and felt pads.

The first stage of developing the prototype involved creating an easy cardboard model to evaluate the exact size it would be. This allowed the team to identify where the components would go and evaluate if it were a sufficient space for the components to fit before committing to a final 3D printed design. After the first layout was completed and reviewed, the frame modeled in SolidWorks and made 3 different iterations. The final chosen frame, motor brackets, and wheels were 3D printed using PETG because it provided a lightweight and durable structure that would be suitable for testing and overall usage.

The assembly first began with mounting the DC motors onto the 3D printed body/shell of the robot by using 3D printed brackets to keep them in place. These brackets made sure the DC motors were secure and guaranteed to stay on. At this point, the next step was attaching a separate plate that held the magnets to make sure it could stay vertically. Which worked to a certain degree; the first few problems from the design started to sprawl out. The magnets' plate attachment exceeded sticking on a vertical surface but couldn't move. This problem occurred around the 6<sup>th</sup> week of the semester. The magnet system would then be a problem for most of the semester and change throughout each prototype iteration.

The second prototype was less of a box shape and more of a shape of a boat, curved edges toward the front side. This prototype incorporated 2 battery sources, one for the Arduino board and the other to power the motors. This prototype iteration also had the addition of a third wheel, a castor wheel to help with balancing the device out. But this design would quickly start failing due to the 3D printed material for the body of the device falling apart and not being able to have the components locked in place inside. The magnets also seemed unreliable in this prototype because they only fit one big magnet and relied on the castor wheel, which also gave the team



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another problem. The castor wheel would scratch and damage the surface it's on and is the reason the team quickly left the idea of a third wheel and gone ahead to investigate a different solution.

This different solution would lead to the fourth prototype that upgraded parts of the earlier prototype. The upgraded parts consisted of a new 3D printed body shell that was now a more open space for the components to fit, and magnets were placed more strategically. After learning how to place the magnets in certain positions and places along the underside of the device, it helped the team understand how the weight and closeness of the magnets to the surface impact the movement of the device the team was developing. This prototype iteration saw more success and having 2 unusual types of magnets placed in certain places along the bottom of the device helped with the overall movement. The power source was not powered from one source and not 2. Cutting a power source helped with the system being able to run from one 12V rechargeable battery source and helped with reducing weight by cutting the extra battery. Powering the device using one power source was done by splicing and soldering 4 wires. Two grounds and two positives. One set would wire into a buck converter, reducing the current and voltage to not fry the system while the other set would connect to the motor/motor driver. The DC motors were able to finally move the device on the board but not to its fullest potential. It could move upward and downward on the board but couldn't move horizontally on the board. Which brings up a different problem in this prototype, the wheels or more specifically the wheel shaft adapter. This shaft adapter helped connect the DC motors with the wheels since they weren't the same size. It was a solution the team thought was resolved but quickly saw negatives of using this shaft adapter. The main reason the shaft adapters were pointed out as problems were because they repeatably snapped and broke whenever there was movement or shifts in direction when the device was moving. This happened right before Week 11, and the team decided to focus and work on everything from scratch throughout Week 11.

The last prototype iteration developed is what the team chose and developed as the final product. After learning and evaluating the earlier prototypes, the team used the lessons learned from magnet placement, power distribution, wheel attachment, and chassis durability to create a more reliable final assembly. The final prototype used a redesigned PETG body with a more open layout so that the Arduino, motor driver, buck converter, battery, sensors, and wiring could fit securely inside the frame. The magnet layout was adjusted so the robot could remain attached to the vertical whiteboard while still allowing the wheels to overcome friction and move. The two-battery system was replaced with a single 12V rechargeable battery and buck converter, which reduced weight and simplified the electrical system. The castor wheel was removed because it scratched the board surface, and the drivetrain was revised to rely on two powered rubber wheels connected to the N20 DC motors. These changes resulted in a final prototype that better balanced adhesion, movement, weight, and cleaning ability. Although the robot still had limitations with consistent horizontal movement and vertical reliability, the final assembly proved the main functions of the autonomous whiteboard cleaner and served as a functional proof of concept for future improvements.

Results and interpretation

Testing proves that the robot satisfied the main aims of the project. The system allows the robot to hook up to a vertical whiteboard with its magnetic attachment system, having enough traction to move. With a motor driven wheel system, the robot shows stable and controlled motion across the board. The navigation system works well with IR sensors to sense edges reliably, preventing the robot from falling off the whiteboard. Obstacle detection is taken care of by the TOF sensor, allowing the robot to move accurately and dynamically. With the aid of a felt pad, a cleaning system can successfully eliminate marker residue during operation. In general, the system achieves the fully autonomous cleaning function and shows reliable performance under typical operating conditions. As the main system works as expected, it was discovered that more features could be added such as color detection and line tracking. These features improve navigation accuracy and enable cleaning patterns to be better optimized, but those characteristics are not necessary for the device's focus. The data confirms that the design is functionally complete to address all key design needs, and future progress on this design should be aimed at expansion, not addressing system deficiencies.

Engineering Analysis of the Autonomous Whiteboard Cleaner

Table: Materials, Cost, and Weight Summary

Electronics

| Component                     | Cost (\$) |
|-------------------------------|-----------|
| Arduino                       | 15.00     |
| Motor Driver                  | 15.00     |
| N20 Motors with Encoders      | 23.99     |
| IR Sensor                     | 5.99      |
| Buck Converter                | 4.90      |
| Battery Pack (1 used)         | 15.00     |
| Screw Terminal Block          | 20.00     |
| Jumper Wires (Male to Female) | 5.00      |
| On/Off Switch (1 used)        | 0.62      |
| Electronics Subtotal          | 107.00    |

Mechanical Components

| Component                             | Cost (\$) |
|---------------------------------------|-----------|
| Wheels (65 mm, rubber tire used)      | 4.50      |
| Magnets                               | 18.00     |
| 3D Printed Frame (Top & Bottom, PETG) | —         |
| 3D Printed Motor Brackets (PETG)      | —         |
| 3D Printed Wheel Rims (PETG)          | —         |
| 3D Printing Material Total            | 2.15      |
| Mechanical Subtotal                   | 24.65     |

Overall Summary

| Category     | Value    |
|--------------|----------|
| Total Cost   | \$131.65 |
| Total Weight | 1.9 lb   |

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Figure 1: Material and Cost Summary

| Given Values                 |                                                                                                                                                                                              |                                                                                  |                                                                    |                                                                                                                                         |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Parameter                    |                                                                                                                                                                                              | Value                                                                            |                                                                    |                                                                                                                                         |
| Robot mass, m                |                                                                                                                                                                                              | 0.855 kg                                                                         |                                                                    |                                                                                                                                         |
| Gravity, g                   |                                                                                                                                                                                              | 9.81 m/s <sup>2</sup>                                                            |                                                                    |                                                                                                                                         |
| Wheel radius, r              |                                                                                                                                                                                              | 0.0325 m                                                                         |                                                                    |                                                                                                                                         |
| Number of motors, n          |                                                                                                                                                                                              | 2                                                                                |                                                                    |                                                                                                                                         |
| Motor torque per motor       |                                                                                                                                                                                              | 1.8 kgf-cm = 0.1765 N-m                                                          |                                                                    |                                                                                                                                         |
| Magnets                      |                                                                                                                                                                                              | 1 × 65 lb and 3 × 70 lb                                                          |                                                                    |                                                                                                                                         |
| Item                         | Equation                                                                                                                                                                                     | Substitution                                                                     | Answer                                                             | Result                                                                                                                                  |
| 1. Robot Weight              | $W = mg$                                                                                                                                                                                     | $W = (0.855) (9.81)$                                                             | $W = 8.39 \text{ N}$                                               | Downward force on the whiteboard                                                                                                        |
| 2. Total Magnetic Force      | $F_{\text{mag}} = F_1 + 3F_2$                                                                                                                                                                | $F_{\text{mag}} = 65 + 3(70) = 275 \text{ lb.}$<br>$F_{\text{mag}} = 275(4.448)$ | $F_{\text{mag}} = 1223.2 \text{ N}$                                | Total rated magnetic pull force                                                                                                         |
| 3. Force Ratio               | $\text{Force Ratio} = F_{\text{mag}} / W$                                                                                                                                                    | $\text{Force Ratio} = 1223.2 / 8.39$                                             | $\text{Force Ratio} = 145.8$                                       | Magnetic force is 145.8 x robot weight                                                                                                  |
| 4. Friction Condition        | $F_f = \mu N$<br>For no sliding: $F_f \geq W$<br>So: $\mu N \geq mg$                                                                                                                         | $\mu N \geq 8.39$                                                                | $\mu N \geq 8.39$                                                  | Friction must be at least 8.39 N                                                                                                        |
| 5. Motor Torque Conversion   | $\tau_{\text{motor}} = 1.8 \text{ kgf-cm}$<br>$1 \text{ kgf-cm} = 0.09807 \text{ N-m}$                                                                                                       | $\tau_{\text{motor}} = 1.8(0.09807)$                                             | $\tau_{\text{motor}} = 0.1765 \text{ N-m}$                         | Torque provided by each motor                                                                                                           |
| 6. Required Torque per Motor | $\tau_{\text{req}} = \frac{mgr}{2}$                                                                                                                                                          | $\tau_{\text{req}} = (0.855) (9.81) (0.0325) / 2$                                | $\tau_{\text{req}} = 0.136 \text{ N-m}$<br>$= 1.39 \text{ kgf-cm}$ | Minimum torque to lift robot vertically                                                                                                 |
| 7. Torque Comparison         | $\text{Torque Ratio} = \tau_{\text{motor}} / \tau_{\text{req}}$                                                                                                                              | $\text{Torque Ratio} = 1.8 / 1.39$                                               | $\text{Torque Ratio} = 1.29$<br>$= 129\%$                          | Each motor exceeds minimum torque                                                                                                       |
| 8. Climbing Condition        | $F_{\text{drive}} = 2\tau / r$                                                                                                                                                               | $F_{\text{drive}} = 2(0.1765) / 0.0325$                                          | $F_{\text{drive}} = 10.86 \text{ N}$                               | 10.86 N > 8.39 N, so lifting is possible in theory                                                                                      |
| 9. Final Conclusion          | $W = 8.39 \text{ N}$<br>$F_{\text{mag}} = 1223.2 \text{ N}$<br>$\tau_{\text{motor}} = 0.1765 \text{ N-m}$<br>$\tau_{\text{req}} = 0.136 \text{ N-m}$<br>$F_{\text{drive}} = 10.86 \text{ N}$ |                                                                                  | Magnets hold and motors can theoretically climb                    | Real-world losses likely caused inconsistent climbing, felt drag, wheel slip, magnetic compression, voltage drop, or drivetrain losses. |

Figure 2: Fundamentals Engineering Analysis

| Battery Capacity and Power Consumption Summary |                                |                                                                 |                                    |                          |
|------------------------------------------------|--------------------------------|-----------------------------------------------------------------|------------------------------------|--------------------------|
| Section                                        | Item                           | Formula                                                         | Calculation                        | Final Answer             |
| Battery Capacity                               | Battery energy                 | $E = V \times Ah$                                               | $E = 12 \times 3$                  | 36 Wh                    |
| Arduino and Sensor Power                       | 5 V current draw               | $I_{5V} = I_{\text{Arduino}} + I_{\text{IR}} + I_{\text{TOF}}$  | $I_{5V} = 50 + 60 + 20 \text{ mA}$ | 130 mA = 0.13 A          |
|                                                | 5 V power                      | $P_{5V} = V \times I$                                           | $P_{5V} = 5 \times 0.13$           | 0.65 W                   |
|                                                | 12 V current for electronics   | $I_{12V, \text{electronics}} = P / (V \times \eta)$             | $I = 0.65 / (12 \times 0.85)$      | approx. 0.064 A          |
| Motor Power                                    | Average load current per motor | Given                                                           | 0.5 A per motor                    | 0.5 A                    |
|                                                | Total motor current            | $I_{\text{motors}} = 2 \times I_{\text{motor}}$                 | $I_{\text{motors}} = 2 \times 0.5$ | 1.0 A                    |
| Total System Current                           | Combined current draw          | $I_{\text{total}} = I_{\text{motors}} + I_{\text{electronics}}$ | $I_{\text{total}} = 1.0 + 0.064$   | 1.064 A                  |
| Battery Life                                   | Ideal battery life             | $t = \text{Battery Capacity} / \text{Total Current}$            | $t = 3 / 1.064$                    | approx. 2.82 hours       |
|                                                | Real-world estimate            |                                                                 |                                    | approx. 2.0 to 2.5 hours |

Figure 3: Power System Analysis

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Figures 1, 2, and 3 show the calculations we used to check if our whiteboard cleaner would work before fully committing to the design. As shown in Figure 1 (Engineering Analysis), the robot weighs about 0.855 kg, which produces a downward force of about 8.39 N. We compared this to the total magnetic force of about 1223 N, which is much higher, meaning the robot should stay attached to a vertical surface. We also checked the motor capability each motor provides about 0.1765 N·m of torque, while only about 0.136 N·m must lift the robot. In theory, this shows the motors are strong enough. The calculated driving force of about 10.86 N also supports that climbing is possible, although real-world losses like friction and inefficiencies still play a role.

On the power side, Figure 2 (Power System Analysis) shows that the total current draw is about 1.064 A, with the motors using most of it (around 1.0 A). Using a 36 Wh battery, this gives an estimated runtime of about 2 to 2.8 hours, which is reasonable for our application. In addition, Figure 3 (Materials and Cost Summary) summarizes the final system weight and total cost based on all selected components. These calculations are important because they helped confirm that our design choices like motor selection, magnet strength, and power setup are valid. They also help explain why some of our earlier prototypes struggled, especially when power or weight distribution was not ideal. Overall, this analysis gave us confidence that the final design should work, while also highlighting areas where real world performance may differ from theory.

### **Design Validation and Modern Engineering Context**

Design validation was conducted through prototyping and testing to ensure that our robot meets all functional requirements. The magnetic adhesion was tested using magnets of several types and sizes. Specifically, we evaluated ferrite magnets and neodymium magnets. Neodymium magnets are significantly stronger than ferrite magnets but are more expensive, depending on their size. We initially tested only neodymium magnets and found that both the cost and magnetic force were too high for our motors, making it difficult for the robot to move. We then tested using only ferrite magnets and found that they were too weak for our application, causing the robot to constantly slip and slide down the whiteboard. Our final solution was to combine both types of magnets. We used two small neodymium magnets placed next to the wheels to increase grip, along with one large ferrite magnet positioned between the wheels to keep alignment along the wheel axis. This design minimized tilting and improved traction. By

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using a combination of both magnet types, we were able to keep costs low while ensuring sufficient magnetic force to prevent the robot from sliding down the board.

Motor choice was tested to meet the robot's torque and space requirements. Initial testing was conducted using standard DC motors rated at approximately 300 RPM; however, these motors were found to be insufficient in torque to overcome the magnetic adhesion and the weight of the robot. To address this issue, a more compact, higher-torque solution was selected: the **N20 gear motor**. The chosen motor is rated at 80 RPM and features a metal integrated gearbox, providing greater durability compared to plastic gearboxes. This lower-speed, higher-torque motor was confirmed through testing, proving its ability to overcome both magnetic adhesion forces and the system's weight while supporting reliable movement. Additionally, its compact form factor allowed for efficient use of space, accommodating other system components within the design constraints.

Sensor functionality was also confirmed, including edge detection to prevent the robot from falling off the board and obstacle detection to find objects in its path. IR sensors were used for edge detection because they were more cost-effective and significantly simpler to integrate into our system compared to using time-of-flight (ToF) sensors on each side. The control logic was designed so that when it was safe to move forward, the sensor output remained at "0." Once an edge was detected, the output changed to "1," signaling the robot to stop and turn around. This logic was confirmed using a dedicated test program that focused solely on sensor performance, allowing the sensors to be manually calibrated to detect the desired distance. In addition, a single ToF sensor was mounted at the front of the robot for obstacle detection. This sensor enabled the robot to identify objects in its path and make decisions such as briefly stopping and navigating around the obstacle, under the assumption that the object was stationary. A separate test program was also developed to figure out the best detection range, ensuring the robot had enough time and space to turn without colliding with the object, while avoiding premature turns. The control logic was confirmed by confirming that the robot could correctly interpret sensor data and execute proper actions, such as turning or stopping when needed. Robot mobility was further evaluated through repeated trials to ensure smooth, consistent, and reliable movement across the whiteboard surface.

Motor driver performance was evaluated through system-level testing. The first design used the **L298N motor driver module**, which functioned adequately during early prototyping; however, as more components were added, a noticeable drop in motor voltage and performance was observed. This was confirmed using a voltage meter, which showed reduced voltage at the motor terminals. The loss was attributed to the L298N's BJT-based design, which introduces a significant voltage drop and limits the effective voltage delivered to the motors, reducing torque

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and efficiency. To address this, the system was updated to use the **TB6612FNG motor driver module**, which utilizes MOSFET-based switching with lower on-resistance, resulting in reduced voltage loss and improved power efficiency. Testing confirmed increased voltage delivery, leading to improved torque output and more consistent overall performance.

### **Cost Analysis BOM (Bill of Materials) (and discussion/explanation)**

The components for the autonomous whiteboard cleaner were selected based on performance, size, and cost, while staying close to the \$100 budget. An Arduino was used for control due to its low cost and compatibility with sensors and motor drivers. The Dual TB6612FNG motor driver was chosen for its compact size and efficient power delivery, improving motor performance while saving space. The system uses N20 motors with encoders for their small size, sufficient torque, and ability to provide feedback for better motor synchronization. For sensing, a combination of IR sensors and a TOF sensor was used to detect edges and obstacles, offering a reliable and affordable alternative to more complex systems.

The power system includes a battery pack and a buck converter to regulate voltage for the Arduino while supplying sufficient power to the motors. Supporting components such as terminal blocks, jumper wires, and an on/off switch were added for safer and more reliable connections. Mechanically, magnets were used for vertical adhesion, with strength adjusted to balance grip and mobility. Rubber wheels with added traction and felt padding improved movement and prevented surface damage. The frame and components were made using 3D-printed PETG parts, allowing for a lightweight and customizable design. The total cost of materials is approximately \$117.65, and the total weight is about 1.9 lb, both of which were used in earlier calculations to confirm the system could work on a vertical surface. Overall, the selected components balance cost, performance, and practicality, resulting in a functional and portable design.

### **Conclusion and Future Work**

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In conclusion, the design and development of the autonomous whiteboard cleaning robot successfully met the primary project objectives, including supporting a cost-effective solution, using magnetic adhesion to remain attached to the whiteboard, and enabling autonomous movement for cleaning. Through iterative prototyping and testing, several challenges were addressed, such as selecting the right magnet configuration to provide sufficient adhesion without causing motor stall, as well as ensuring adequate torque for reliable movement without slippage. Additionally, the integration of higher-torque motors and an efficient motor driver improved traction and minimized voltage losses within the system. Overall, the project provides an effective and practical approach to autonomous whiteboard cleaning.

While the robot meets the current design requirements, several improvements could enhance its performance and functionality. One potential upgrade is the integration of the wheel encoders included with the motors, which would allow the robot to track its position and map the whiteboard surface more accurately. This would enable more efficient and structured cleaning paths. Additionally, incorporating an IR or color-detection sensor could allow the robot to distinguish between different marker shades, enabling it to find target areas that require cleaning rather than moving blindly across the surface. Other potential improvements include optimizing the control algorithm for better path planning, implementing closed-loop motor control for more accurate speed and position regulation, and refining the mechanical design to further reduce weight and increase energy efficiency. These enhancements would move the system closer to a fully autonomous and intelligent cleaning solution.

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## Appendix